

Are Big Cities Important for Economic Growth?

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Are cities “Engines of growth?”

- ▶ US output per hour increased by a factor of 10.3 from 1900-2010, an annual growth rate of 2.12%. About 70% is due to TFP growth, most of the rest is increases in human capital per capita.
- ▶ The literature discusses (and estimates) four mechanisms by which cities affect the level and growth of output,
 - ▶ Urban scale economies for output,
 - ▶ Urban scale economies for innovation,
 - ▶ Urban scale economies for human capital accumulation,
 - ▶ Sorting of people to more productive cities.
- ▶ We ask whether the first three made an important contribution to the growth in TFP. What would have happened if city size were capped at 1m or 100k?
- ▶ ‘Sorting’ is orthogonal to our argument, but the literature suggests it is small, and (we think) its empirical foundations are suspect.
- ▶ NB: unmeasured human capital is TFP.

Literature I

- ▶ *Sorting of people to more productive cities*: Hsieh and Moretti [2019], Fajgelbaum and Gaubert [2020], Rossi-Hansberg et al. [2019], Duranton and Puga [2023]. Finding an important effect depends on whether cross-city transfers are allowed. TFP is (partly) in the soil of each city – it is not (wholly) in the people.
- ▶ *Urban scale economies for output*: Fajgelbaum and Gaubert [2020], Rossi-Hansberg et al. [2019], Duranton and Puga [2023], city growth increases output. Davis et al. [2014] density $\implies$ income $\implies$ investment $\implies$ density $\implies$ Urban scale economies account for about 10% of TFP growth in late 20th century US.
- ▶ *Urban scale economies for innovation*: Desmet and Rossi-Hansberg [2014], only qualitative predictions.

Literature II

- ▶ *Urban scale economies for human capital accumulation*: Black and Henderson [1999] (just theory), Duranton and Puga [2023] density $\implies$ income $\implies$ human capital investment $\implies$ TFP growth. Same basic argument as Black and Henderson [1999] and Davis et al. [2014].

What we do:

- ▶ Urban scale affects the level of output and human capital. Parameters from the literature.
- ▶ Urban scale affects the rate of innovation directly. Parameters from the literature. This is either (1) correct, or (2), a reduced form of more complicated mechanisms in e.g. Davis et al. [2014] and Duranton and Puga [2023].
- ▶ Constrain relationship between research effort and TFP growth to decline with income as we observe.

Importance of urban scale economies for output, σ_A

$Y_t, Y_{it} \sim$ Total output, output of city i , period t

$L_t, L_{it} \sim$ Total population, population of city i , in period t

$A_{it} = A L_{it}^{\sigma_A} \sim$ TFP of city i in period t

$$Y_{it} = (A_{it} L_{it}^{\sigma_A}) L_{it}$$

To make math easy, $L_{it} = L_{jt}$, all i, j , each t and
 $L_{i,1900} = \kappa L_{i,2010}$, then

$$\begin{aligned} \frac{Y_{1900}}{L_{1900}} &= A(L_{i,1900})^{\sigma_A} \\ \frac{10 Y_{1900}}{L_{2010}} &= A(\kappa L_{i,1900})^{\sigma_A} \end{aligned}$$

This means $\kappa = 10^{1/\sigma_A} \approx 10^{20}$. Urban scale economies for output are too small to play an important role in the growth of TFP.

Importance of urban scale economies for innovation, σ^B I

$P_{it} \sim$ Patents in city i in period t

$L_{it} \sim$ Population of city i in period t

$\sigma_B \sim$ patent elasticity of city size.

$$P_{it} = (BL_{it}^{\sigma_B})L_{it}$$

Assume $L_{it} = L_{jt}$, all i, j . What happens if we double the size of all cities, holding population constant and $\sigma_B = 0.06$?

$$\frac{P_t^1}{P_t^0} = \frac{\sum_1^{137} B(2L_i)^{1+\sigma_B}}{\sum_1^{274} B(L_i)^{1+\sigma_B}} = 2^{\sigma_B} \approx 1.04$$

If the growth rate is proportional to patents, this reduces the growth rate from 0.018 to $\frac{0.0212}{1.04} = 0.0204$. Applied to 1900-2010, output in 2010 is 9.4 times 1900, instead of 10.3.

Importance of urban scale economies for innovation, σ^B II

- ▶ This suggests that the urban scale effect for innovation need not cumulate to large effects on output.
- ▶ However, σ_B could be as large as 0.2, and many cities grew by more than a factor of two. So this example does not seem conclusive.

Importance of urban scale economies for innovation, σ^B III

- ▶ So far, we assume that the growth rate is proportional to patenting. This is not what we observe.
- ▶ Let $\frac{\Delta Y_{t+1}}{Y_t}$ denote the growth rate of output. (Loosely) Bloom et al. [2020] estimate that

$$\frac{\Delta Y_{t+1}}{Y_t} = \frac{P_t}{\left(\frac{Y_t}{L_t}\right)^{3.1}}$$

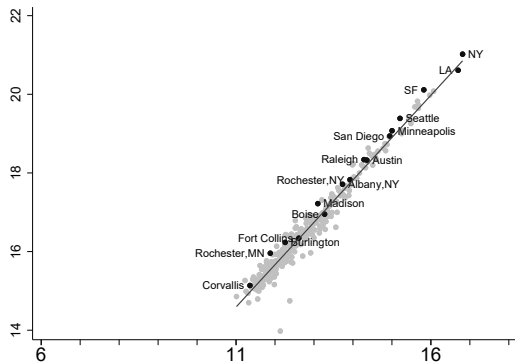
With $\left(\frac{Y_{2010}}{L_{2010}}\right) = 10 \left(\frac{Y_{1900}}{L_{1900}}\right)$ the contribution of a marginal patent to growth in 2010 is about $\frac{1}{1400}$ of its contribution in 1900.

- ▶ Urban scale economies for innovation probably do not have a large effect on the growth of TFP.
- ▶ The paper is about making this argument more precisely.

Data

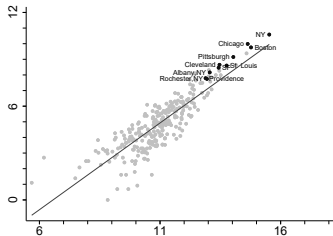
- ▶ 274 2010 MSAs plus “rural”.
- ▶ County level population, 1900-2020.
- ▶ BEA County level output data, 2010.
- ▶ CUSP patent data, 1900-2014. Date and county of residence for inventor, for all US patents.

$\ln(\text{output})$ vs. $\ln(\text{population})$ in 2010

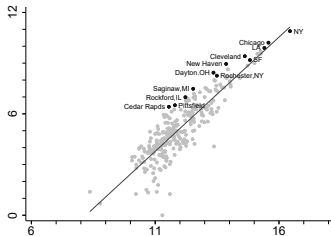


Slope = 1.08

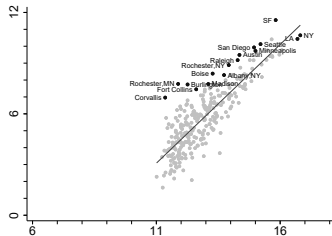
$\ln(\text{patents})$ vs. $\ln(\text{population})$.



1900 - Slope = 1.11



1950 - Slope = 1.35



2010 - Slope = 1.41

City level production of output

- Production function for MSA i

$$Y_{it} = A_{it} (K_{it})^{\gamma} \left(h_{it}^Y \ell_{it}^Y L_{it} \right)^{1-\gamma}$$

$A_{it} \sim$ is city-level TFP for output. $h_{it}^Y \sim$ human capital for output. $\ell_{it}^Y \sim$ share labor for output.

- TFP is

$$A_{it} = \bar{A}_t \tilde{A}_{it} \hat{A}_{it}$$

$\bar{A}_t \sim$ time specific national component

$\tilde{A}_{it} = L_{it}^{\sigma_A} \sim$ the urban scale effect for output

$\hat{A}_{it} \sim$ a city-decade specific idiosyncratic term (sorting)

- Our paper is about $\bar{A}_t$ and $\tilde{A}_{it}$. We take $\hat{A}_{it}$ as given and invariant in our counterfactuals.

National production of output

- Assume that physical capital is freely mobile

$$\implies \frac{Y_{it}}{K_{it}} = \frac{Y_t}{K_t} \quad \forall i$$

- Substitute into

$$Y_{it} = A_{it} (K_{it})^\gamma \left(h_{it}^Y \ell_{it}^Y L_{it} \right)^{1-\gamma}$$

to eliminate K_{it} , and then sum across cities,

$$Y_t = \left(\frac{K_t}{Y_t} \right)^{\gamma/(1-\gamma)} \sum_i A_{it}^{1/(1-\gamma)} h_{it}^Y \ell_{it}^Y L_{it}.$$

Base vs. Alternative cases

- ▶ We observe the *Base Case*.
- ▶ In the *Alternative Case* we cap city size at 1m or 100k.
- ▶ Assume $\hat{A}_{it}$ (idiosyncratic component of city TFP), h_{it}^Y (human capital), and ℓ_{it}^Y (labor share for output) are the same in both cases.
- ▶ Assume that the aggregate ratio of capital to output, K/Y is the same in both cases (follows from fixed investment rate on Balanced Growth Path.)
- ▶ Base and Alternative cases differ in values of $\bar{A}_t$ (national component), $\tilde{A}_{it}$ (scale economies), and city populations.

Alternative relative to Base

- Recall that output is given by,

$$Y_t = \left(\frac{K_t}{Y_t} \right)^{\gamma/1-\gamma} \sum_i A_{it}^{1/(1-\gamma)} h_{it}^Y \ell_{it}^Y L_{it}.$$

- So ratio of Alternative to Base is,

$$\frac{Y_t^{\text{alt}}}{Y_t^{\text{base}}} = \frac{\sum_i (A_{it}^{\text{alt}})^{1/(1-\gamma)} h_{it}^Y \ell_{it}^Y L_{it}}{\sum_i (A_{it}^{\text{base}})^{1/(1-\gamma)} h_{it}^Y \ell_{it}^Y L_{it}}$$

- Do a bunch of algebra.....

$$\frac{Y_t^{\text{alt}}}{Y_t^{\text{base}}} = \sum_i \frac{Y_{it}^{\text{base}}}{Y_t^{\text{base}}} \left(\frac{A_{it}^{\text{alt}}}{A_{it}^{\text{base}}} \right)^{1/(1-\gamma)}$$

Implementing the counterfactual

- Keep city populations constant.
- Assign every city the level of $\tilde{A}_{it}$ that it would have if city sizes were constrained to L_{max} .

$$\frac{\tilde{A}_{it}^{alt}}{\tilde{A}_{it}^{base}} = \min \left(1, \left(\frac{L_{max}}{L_{it}} \right)^{\sigma_A} \right)$$

This is just “turning down” scale economies for output.

- The ratio of Alternative to Base output becomes,

$$\frac{Y_t^{alt}}{Y_t^{base}} = \sum_i \frac{Y_{it}^{base}}{Y_t^{base}} \min \left(1, \left(\frac{L_{max}}{L_{it}} \right)^{\sigma_A/(1-\gamma)} \right)$$

Notice the magic

- Ratio of Alternative to Base output (for reference)

$$\frac{Y_t^{\text{alt}}}{Y_t^{\text{base}}} = \sum_i \frac{Y_{it}^{\text{base}}}{Y_t^{\text{base}}} \min \left(1, \left(\frac{L_{\max}}{L_{it}} \right)^{\sigma_A/(1-\gamma)} \right)$$

- We don't have to observe city-specific factors of production (human and physical capital).
- We also don't have to try to estimate city-time specific productivity $\hat{A}_{it}$.
- All we need is city output (Y_{it}^{base}), city population (L_{it}) and the value of σ_A .

Human capital extension

- ▶ Human capital, h_{it}^Y , is a function of city-decade specific inputs (years of education and their quality), S_{it} and ϕ_{it} , according to

$$h_{it}^Y = \exp(\phi_{it} S_{it}) L_{it}^{\sigma_H},$$

- ▶ σ_H measures scale economies for human capital accumulation.
- ▶ With endogenous human capital, swap $\frac{\sigma_A}{1-\gamma}$ for $\frac{\sigma_A}{1-\gamma} + \sigma_H$ in the expression for $\frac{Y_t^{\text{alt}}}{Y_t^{\text{base}}}$. This gives,

$$\frac{Y_t^{\text{alt}}}{Y_t^{\text{base}}} = \sum_i \frac{Y_{it}^{\text{base}}}{Y_t^{\text{base}}} \min \left(1, \left(\frac{L_{\max}}{L_{it}} \right)^{\frac{\sigma_A}{1-\gamma} + \sigma_H} \right)$$

- ▶ This is the expression we use for our analysis of scale economies for output and human capital accumulation.

Estimates of scale economies for output and human capital

σ_A	σ_H	$\frac{\sigma_A}{1-\gamma} + \sigma_H$	Source	Data
3.4%	~ 0	5.1%	Ciccone & Hall (1996)	US, State output, 1988
2.5%	$\equiv 0$	3.7%	Combes et al. (2008)	French, Ind. wages, 1976-98
1.5%	2.9%	5.1%	delàRoca & Puga (2017)	Spanish, Ind. wages, 2004-9
2.9%	3.1%	7.6%	Duranton & Puga (2023)	US, Ind. wages, ca. 1979-2020
2.7%	$\equiv 0$	4.1%	Glaeser & Gottlieb (2008)	US, Ind. wages, 2000
8.6%	$\equiv 0$	13%	Glaeser & Gottlieb (2009)	US, MSA output, 2000

- We consider values of $\frac{\sigma_A}{1-\gamma} + \sigma_H$ of 0.04, 0.08, and 0.12.

Ratio of Alternative to Base output for 2010

$\frac{\sigma_A}{1-\gamma} + \sigma_H$	$L_{max} = 1m$	$L_{max} = 100k$
0.04	0.94	0.84
0.08	0.88	0.72
0.12	0.83	0.62

Each table cell reports this calculation for given parameter values and BLS data for MSA level output in 2010.

$$\frac{Y_t^{\text{alt}}}{Y_t^{\text{base}}} = \sum_i \frac{Y_{it}^{\text{base}}}{Y_t^{\text{base}}} \min \left(1, \left(\frac{L_{max}}{L_{it}} \right)^{\frac{\sigma_A}{1-\gamma} + \sigma_H} \right)$$

Production of patents I

- ▶ City size affects research productivity and hence TFP growth. This shows up in the city invariant component of TFP, $\bar{A}_t$.
- ▶ Use patenting as a measure of research productivity (within a cross-section). $P_{it} \sim$ count patents. $h_{it}^R \sim$ human capital per research worker. $\ell_{it}^R \sim$ share of population working in research (no capital).

$$P_t = \sum_i B_{it} h_{it}^R \ell_{it}^R L_{it}.$$

- ▶ $B_{it} = \hat{B}_{it} \bar{B}_t \tilde{B}_{it}$ where: $\hat{B}_{it} \sim$ city-decade idiosyncratic term; $\bar{B}_t \sim$ time-specific national component; $\tilde{B}_{it} = L_{it}^{\sigma_B} \sim$ scale effect for innovation. (Same TFP structure as for output).

Production of patents II

- ▶ The production of patents has the same structure as the production of outputs (save for the omission of capital).
- ▶ The ratio of patents in the Alternative to the Base case is

$$\frac{P_t^{\text{alt}}}{P_t^{\text{base}}} = \sum_i \frac{P_{it}^{\text{base}}}{P_t^{\text{base}}} \min \left(1, \left(\frac{L_{\max}}{L_{it}} \right)^{\sigma_B} \right)$$

- ▶ We need only city patents, P_{it} , city population, L_{it} , and σ_B .
- ▶ Moretti (2021) $\implies \sigma_B = 0.06$ (panel data).
- ▶ Carlino et al (2007) $\implies \sigma_B = 0.20$ (cross-section).

Ratio of Alternative to Base 2010 patents		
σ_B	$L_{max} = 1m$	$L_{max} = 100k$
0.06	0.93	0.83
0.20	0.82	0.58

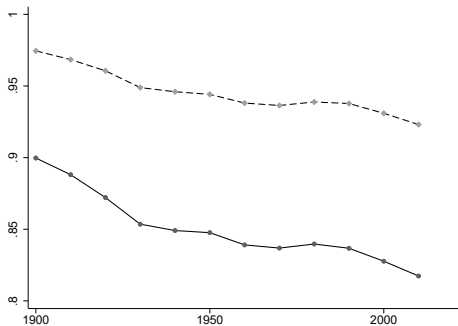


Figure: L_{max} is 1m (diamonds), 100k (squares). $\sigma_B = 0.06$.

Three comments

- ▶ No particular relationship between city-specific productivity in producing output (the A terms) and city-specific productivity in producing research (the B terms). Some places might be good for one but not the other.
- ▶ No particular relationship between human capital used to produce output, on the one hand, and the inputs used to produce research, on the other.
- ▶ Our methodology is conservative. If city population is divided into two daughter cities, the number of researchers in each daughter city is half the original. If researchers were not divided evenly, the reduction in R&D output would be smaller than what we estimate.

From Patents to TFP growth

- ▶ From Jones (2002), Bloom et al (2020)

$$\frac{\Delta \bar{A}_t}{\bar{A}_t} = \alpha \frac{S_t^\lambda}{\bar{A}_t^\beta}$$

$S \sim$ number of researchers – *new notation – not something we measure.*

- ▶ λ measures decreasing returns to scale research effort within a period, e.g. Leibniz and Newton.
- ▶ β measures “fishing out”, a PhD takes six years in the 2020s instead of three in the 1960s.
- ▶ Bloom et al. values for λ and β :
 - ▶ Primary: $\lambda = 1$ and $\beta = 3.1$
 - ▶ Alternative: $\lambda = 3/4$ and $\beta = 2.4$
 - ▶ Naive: $\lambda = 1$ and $\beta = 0$

Adapting to our setting

- ▶ Allow TFP growth to depend on scale economies for innovation. S_t becomes R_t “efficiency units of research effort”.
- ▶ Allow for explicit time-varying term η_t in equation that relates TFP growth to effective research effort (technical assumption).
- ▶ This gives,

$$\frac{\Delta \bar{A}_t}{\bar{A}_t} = \alpha \eta_t \frac{R_t^\lambda}{\bar{A}_t^\beta}$$

Recipe for Alternative path of TFP growth

- Impute $(R_{1900}^{\text{base}}, \dots, R_{2010}^{\text{base}})$ from observed $(\bar{A}_{1900}^{\text{base}}, \dots, \bar{A}_{2010}^{\text{base}})$ (from Gordon) using the Bloom et al (2020) relationship,

$$\frac{\Delta \bar{A}_t}{\bar{A}_t} = \alpha \eta_t \frac{R_t^\lambda}{\bar{A}_t^\beta}.$$

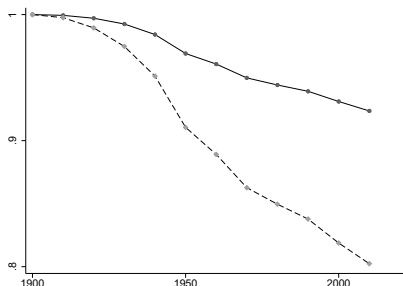
- Allocate research effort R_t^{base} to cities in proportion to observed share of patents P_t^{base} .
- Adjust R_{it}^{base} to R_{it}^{alt} by assuming

$$\frac{R_{it}^{\text{alt}}}{R_{it}^{\text{base}}} \propto \frac{P_{it}^{\text{alt}}}{P_{it}^{\text{base}}}$$

- Sum over cities to get R_t^{alt} .
- Use Bloom et al. again to get $(\bar{A}_{1900}^{\text{alt}}, \dots, \bar{A}_{2010}^{\text{alt}})$.

$$\bar{A}_{\text{alt}} / \bar{A}_{\text{base}}$$

Parameters	$L_{\max} = 1m$	
	$\sigma_B = .06$	$\sigma_B = .20$
$\lambda = 1$ and $\beta = 3.1$	0.979	0.935
$\lambda = .75$ and $\beta = 2.4$	0.980	0.939
$\lambda = 1$ and $\beta = 0$	0.924	0.790



$L_{\max} = 1m$ (squares) and 100k (diamonds). $\sigma_B = 0.06$, $\lambda = 1$, and $\beta = 0$.

Combining scale economies for output and innovation

Output relative to 2010 baseline

$\frac{\sigma_A}{1-\gamma} + \sigma_H$	$\sigma_B = 0.06$			$\sigma_B = 0.20$		
	0.04	0.08	0.12	0.04	0.08	0.12
$\lambda = 1.00, \beta = 3.1$	0.917	0.863	0.816	0.877	0.825	0.780
$\lambda = 0.75, \beta = 2.4$	0.919	0.865	0.817	0.880	0.829	0.783
$\lambda = 1, \beta = 0$	0.866	0.815	0.770	0.741	0.697	0.658

$$L_{max} = 1m$$

$\sigma_A \sim$ scale economies for output

$\sigma_H \sim$ scale economies for human capital

$\sigma_B \sim$ scale economies for innovation

$\lambda \sim$ DRS for research effort

$\beta \sim$ “fishing out”

$\gamma \sim$ capital share

Cities as engines of growth?

- ▶ US GDP per hour of work increased by a factor of 10.3 between 1900 and 2010. This is 2.12% per year.
- ▶ Using our base case parameters, limiting city size to 1 million would mean that 2010 output would be 92% of observed value.
- ▶ Growth would have been 1.97% per year.
- ▶ For most extreme plausible parameters, growth would have been 1.65% per year.
- ▶ Restricting city sizes to 1m involves moving about half the population. This is a catastrophic reorganization.
- ▶ What about sorting?

Conclusion

Either:

- ▶ Cities make a marginal contribution to economic growth. Or...
- ▶ Existing estimates of urban scale economies for output are way off. Or...
- ▶ Existing estimates of urban scale economies for R&D productivity are way off. Or...
- ▶ Existing estimates of the mapping from R&D to TFP growth are way off.

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